

1 Causes of variation in vaccine efficacy in a wild rodent
2 population

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TODO LIST

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13 **Individuals vary broadly in their response to vaccination and subsequent chal-**
14 **lenge infection, with poor vaccine responders causing persistence of both**
15 **infection and transmission in populations.**

16 **Keywords:** Helminth; vaccine; immunoparasitology; machine learning; pathway ana-
17 lysis; ageing.

OUTLINE

18 Two main classes of vaccination: [AD, DA] vs [DD]. DD is the booster (can we rescue
19 poor responders?). We are focussing on the variation within vaccine regimens: why do
20 individuals vary despite identical vaccines?

21 Aims:

- 22 • what drives variation in immune response?
- 23 • why do vaccine lose efficacy in the wild?
- 24 • can we recover the non-responders, with boosters, diet, other interventions...
- 25 • vaccination to probe wild immune systems
- 26 • methods development

27 1. Partition differences between adjuvant and dose regimen

- 28 • Be clear that we chose alum to elicit type 2 antibody responses, because we can
29 really only measure IgG1 in this system. Discussion: we might see different
30 relationships between diet and antibody response if we measure other antibody
31 isotypes.
- 32 • Comparison between AD, DA, DD

33 2. Total effects of interventions

- 34 • $OD \sim V + W + D + V : W + V : D + W : D$ on all mice => plot of OD by Diet
35 (D) and Lab vs Wild (W)
- 36 • $OD \sim V + P + D + V : P + V : D + P : D$ on wild mice only
- 37 • Depending on the results of the models above, break down further.

- 38 3. Disentangling effects of environmental covariates with SCM: how do Sex, Weight,
39 Parasite infection, reproductive status, and time since vaccination affect immunisation
40 efficacy?

INTRODUCTION

41 Individuals vary widely in their responses to vaccination and little is known about the
42 causes underlying low vaccine efficacy, nor how effective a vaccine will be “in the real
43 world” against the pathogen for which it has been developed.¹

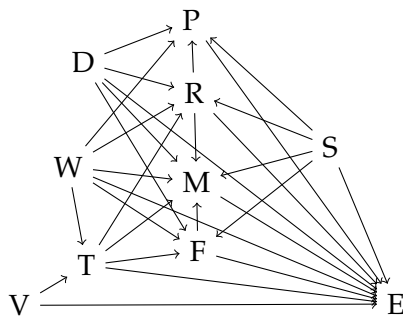
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RESULTS

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METHODS

71 Notebooks produced using Pluto.jl v.0.17.7²

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COMPETING INTERESTS

72 The authors declare that there are no competing interests.

AUTHOR CONTRIBUTIONS

REFERENCES

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FIGURE LEGENDS

SUPPLEMENTARY INFORMATION

⁷⁷ **Supplementary Data 1**

⁷⁸ **Supplementary Data 2**

⁷⁹ **Supplementary Data 3**

⁸⁰ **Supplementary Figures**